ASTA607: COMPUTATIONAL STATISTICS I

Q.1

a) describing the dataset

```
?crabs


Description

The crabs data frame has 200 rows and 8 columns, describing 5
morphological measurements on 50 crabs each of two colour forms
and both sexes, of the species Leptograpsus variegatus collected
at Fremantle, W. Australia
```

b) loading the dataset and displaying its structure

```
data("crabs")

 str(crabs)
 'data.frame':	200 obs. of  8 variables:
  $ sp   : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1
 1 ...
  $ sex  : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2
 2 ...
  $ index: int  1 2 3 4 5 6 7 8 9 10 ...
  $ FL   : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8
 11.8 ...
  $ RW   : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
  $ CL   : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2
 25.2 ...
  $ CW   : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8
 29.3 ...
  $ BD   : num  7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3
 ...
```

c) Frequency table for species and sex

```
Frequency_SS <- table(crabs$sp,crabs$sex)
> Frequency_SS

     F  M
  B 50 50
  O 50 50
```

d) constructing bar chart

```
par(mar=c(4,4,1,1))
> barplot(Frequency_SS,xlab = "crabs",ylab = 'Frequency', col=2:
4 , main = "Morphological Measurements on Leptograpsus Crabs")
```

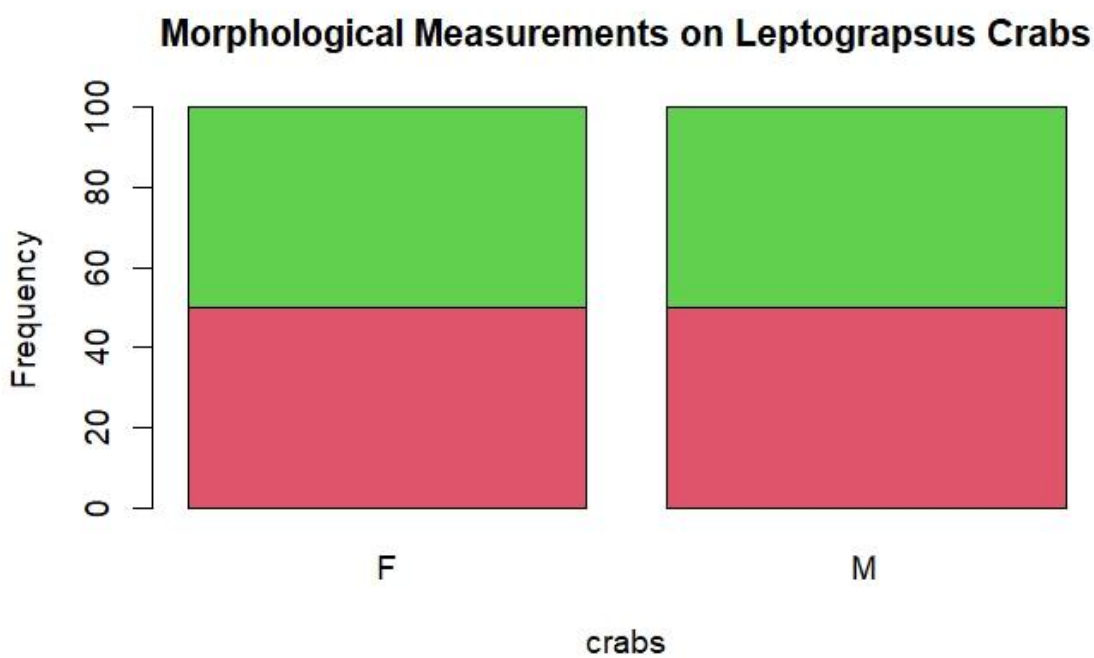

Figure 1:bar chart summarizing the group composition.

From the above figure , the morphological measurements on the leptograpus crabs. The diagram show the balance across the groups. Each of two colour forms and both sexes, of the species.

Q2 a) producing a box plot

```
boxplot(crabs$BD~crabs$sp, xlab = 'species',ylab = "BD" , m
ain= "Distribution of body depth by species", col= rainbow(
3), )
```

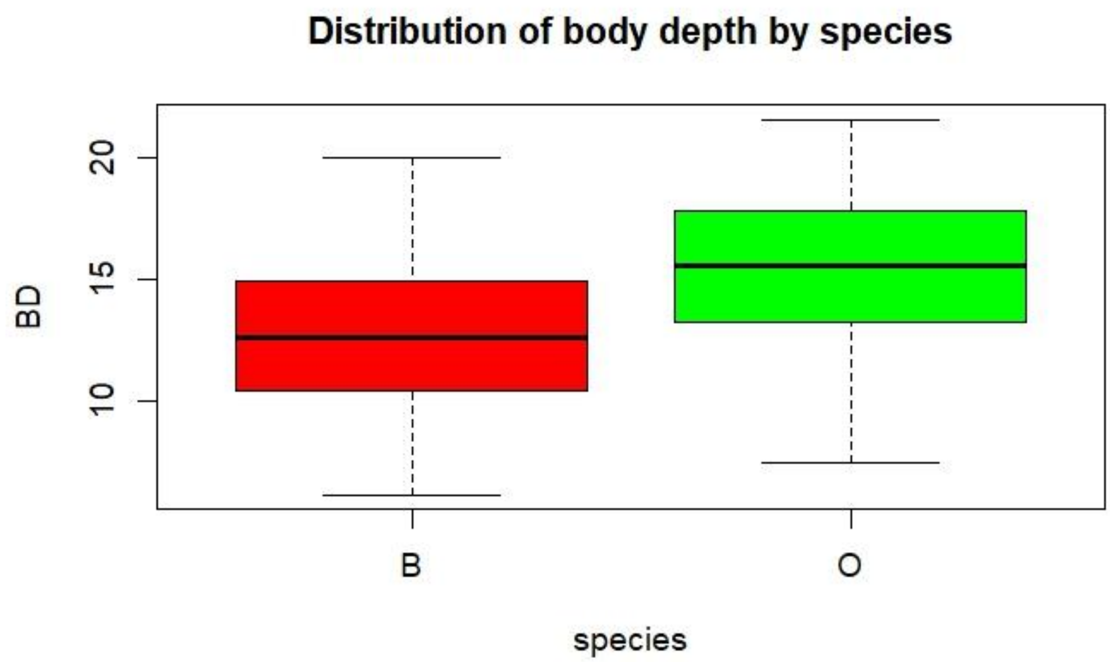

Figure 2: boxplots showing the distribution of body depth.

The B species is equally distributed. The O species has a outlier .

b) central tendency between the species and sex

```
CT <- crabs%>% group_by(sp)%>%summarise(mean=mean(BD),median=med
ian(BD))
> CT
# A tibble: 2 × 3
  sp     mean median
  <fct> <dbl>  <dbl>
1 B      12.6   12.6
```

```
2 O       15.5   15.6

CTsex <- crabs%>% group_by(sex)%>%summarise(mean=mean(BD),median
=median(BD))
> CTsex
# A tibble: 2 × 3
  sex    mean median
  <fct> <dbl>  <dbl>
1 F       13.7   13.8
2 M       14.3   14.2
```

Q 3

a) Producing a histogram

```
par(mfrow=c(1,2))
> hist(subset(crabs,sp=="B")$CL, xlab="B", ylab="Frequenc
y", col=2, main="Carapace length distribution by Species"
)
> hist(subset(crabs,sp=="O")$CL, xlab="O", ylab="Frequenc
y", col=3, main = "Carapace length distribution by Specie
s")
```

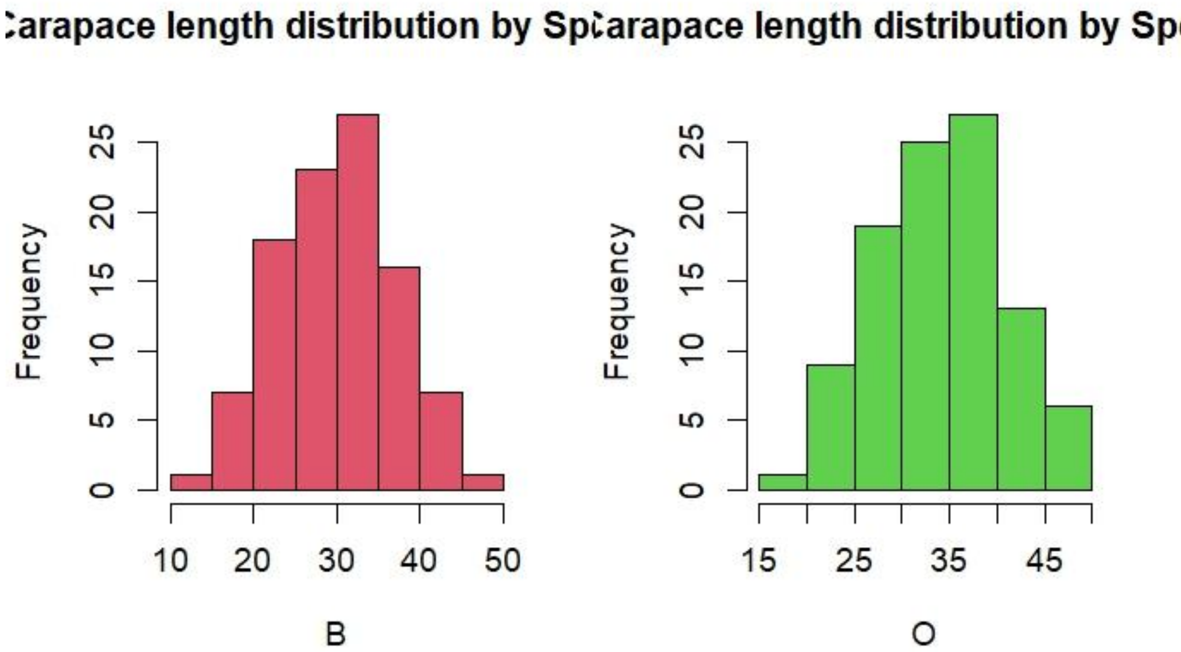

Figure 3: histogram of carapace length for each species.

From the above diagram, comparing the carapace length distribution by species shows that the species B is symmetrically distributed there are no outliers as compared to the negatively skewed O species.

Q4. a) computing the various means

```
MM_FL <- mean(crabs$FL)
MM_RW <- mean(crabs$RW)
MM_CL <- mean(crabs$CL)
MM_CW <- mean(crabs$CW)
MM_BD <- mean(crabs$BD)

Prints:
[1] 15.583
> MM_RW
[1] 12.7385
> MM_CL
[1] 32.1055
> MM_CW
[1] 36.4145
> MM_BD
[1] 14.0305
```

b) Constructing a grouped bar chart

```
par(mfrow= c(1,5))
par(mfrow=c(1,5) )
barplot(MM_FL, xlab = 'FL', ylab = 'means', col = 2)
barplot(MM_RW, xlab = 'RW', ylab = 'means', col = 3)
barplot(MM_CL, xlab = 'CL', ylab = 'means', col = 4)
barplot(MM_CW, xlab = 'CW', ylab = 'means', col = 5)
barplot(MM_BD, xlab = 'BD', ylab = 'means', col = 6)
```

c)

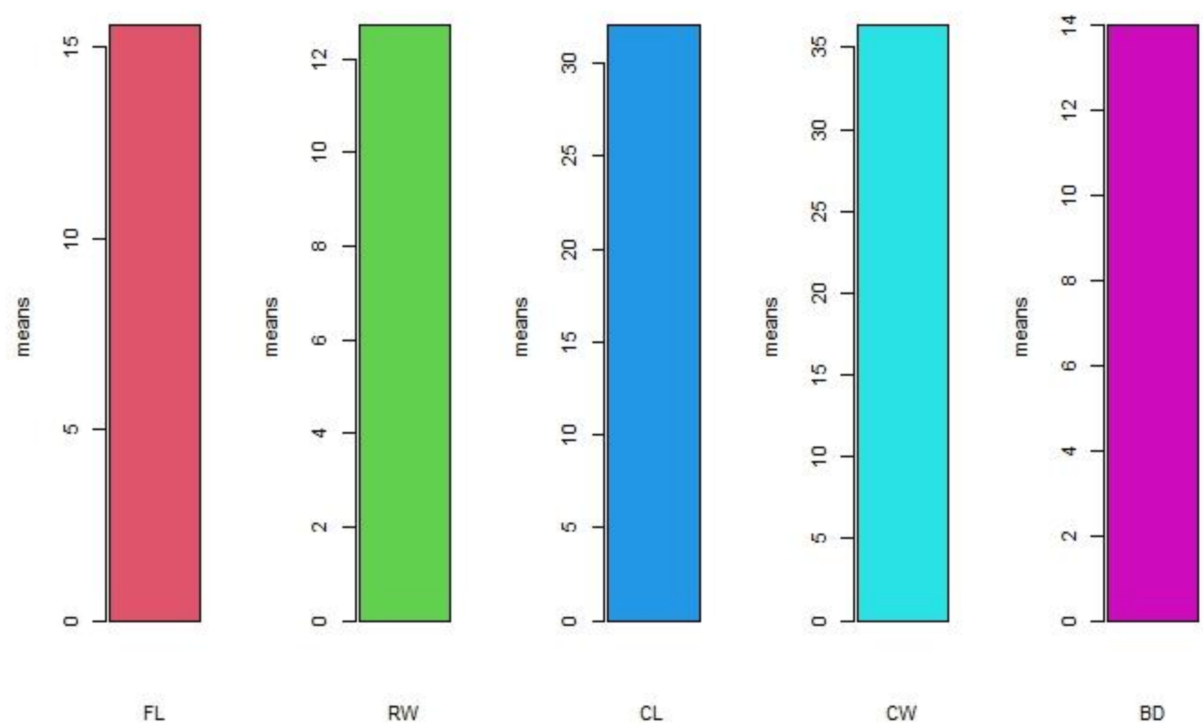

Figure 4:group bar chart comparing of means.

Q.5

```
cleanNA<- na.omit(crabs)


## visualizing
pairs(cleanNA,  main = "Scatterplot Matrix of the five  measurements
of the crabs dataset ", col= rainbow(7))
```

Figure 5

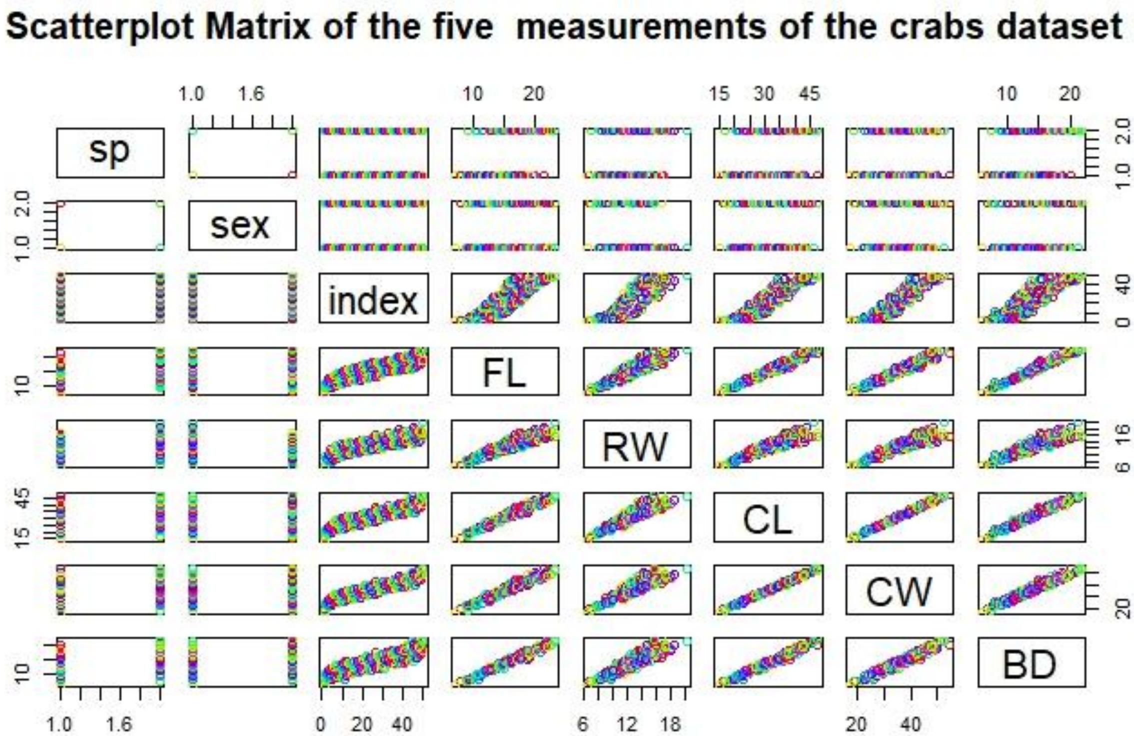

Figure 6: scatterplot of the crabs' data set

There is a strong positive correlation of the relationship among variables. It seems to be a linear relationship.